Permutations that preserve persymmetry

Understanding when P^TAP is persymmetric for every persymmetric A

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1. What is the question?

Find the permutation matrices P for which

$$A \text{ persymmetric} \quad \Longrightarrow \quad P^TAP \text{ persymmetric}$$

for *every* real persymmetric matrix A of a fixed size n. Here P is chosen independently of A. The answer is that the permutation must preserve the mirror pairs of indices. We begin with actual row and column operations, then prove the complete classification.

2. What does P^TAP actually do?

A permutation matrix contains exactly one 1 in each row and each column, with all remaining entries zero. Its columns are a reordering of the standard basis vectors, so $P^TP = I_n$ and $P^{-1} = P^T$.

Fix the following convention. An ordering $p = (p(1), \ldots, p(n))$ determines P by taking column i of P to be $e_{p(i)}$. Then

$$B = P^TAP \quad \Longrightarrow \quad b_{ij} = a_{p(i),p(j)}. \tag{1}$$

In words, reorder the columns of A according to p, and then reorder its rows according to the same list. This is a similarity transformation.

A valid example: reverse the order

Take

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 2 \\ 6 & 4 & 1 \end{pmatrix}$$

and the ordering $p = (3, 2, 1)$, so $P = J_3$. Right multiplication reorders the columns:

$$AP = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 5 & 4 \\ 1 & 4 & 6 \end{pmatrix}.$$

Left multiplication by P^T then reorders the rows:

$$P^TAP = \begin{pmatrix} 1 & 4 & 6 \\ 2 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}.$$

The three pairs of reflected entries are now 1 and 1, 4 and 4, and 2 and 2. The matrix remains persymmetric.

3. The mirror-pair rule

Let $r(i) = n + 1 - i$ be reversal of an index. The mirror pairs are

$$\{1, n\}, \quad \{2, n-1\}, \quad \ldots$$

If n is odd, the middle index $(n+1)/2$ is its own partner. The permitted operations are:

1. Move a whole mirror pair to another mirror pair.
2. Independently swap the two members within any pair.
3. Leave the middle index fixed when n is odd.

The numerical test for the ordering in (1) is

$$\boxed{p(n+1-i) = n + 1 - p(i) \quad (1 \le i \le n).} \tag{2}$$

Equivalently, opposite entries of the ordering list must add to $n + 1$.

A four-dimensional worked example

For $n = 4$, the pairs are $\{1, 4\}$ and $\{2, 3\}$. The ordering $p = (2, 1, 4, 3)$ is valid because $2 + 3 = 5$ and $1 + 4 = 5$. Its matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

With

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 3 \\ 8 & 9 & 6 & 2 \\ 10 & 8 & 5 & 1 \end{pmatrix},$$

reordering both rows and columns gives

$$P^T A P = \begin{pmatrix} 6 & 5 & 3 & 7 \\ 2 & 1 & 4 & 3 \\ 8 & 10 & 1 & 5 \\ 9 & 8 & 2 & 6 \end{pmatrix}.$$

For example, the new $(1, 2)$ and $(3, 4)$ entries are both 5; the new $(2, 1)$ and $(4, 3)$ entries are both 2. All reflected pairs agree.

By contrast, $q = (2, 1, 3, 4)$ is invalid: its opposite entries 2 and 4 sum to 6, not 5. The original pair $\{1, 4\}$ has been split.

An odd-dimensional example

For $n = 5$, the ordering $p = (2, 5, 3, 1, 4)$ is valid:

$$p(1) + p(5) = 2 + 4 = 6, \quad p(2) + p(4) = 5 + 1 = 6, \quad p(3) = 3.$$

The ordering $(2, 5, 1, 3, 4)$ is invalid; it moves the middle index and breaks the second pair. No matrix multiplication is needed to decide this.

4. The complete characterization and proof

Theorem 1. *Let P be an $n \times n$ permutation matrix, with ordering p as defined in* (1)*. The following statements are equivalent:*

1. *$P^T AP$ is persymmetric for every real persymmetric A.*
2. *$PJ_n = J_n P$.*
3. *$P^T J_n P = J_n$.*
4. *$p(n+1-i) = n+1-p(i)$ for every i.*

Proof:

Suppose $PJ_n = J_n P$. Transposing this equality gives $J_n P^T = P^T J_n$. Let $A = J_n A^T J_n$ and set $B = P^T AP$. Then

$$\begin{aligned} J_n B^T J_n &= J_n (P^T AP)^T J_n \\ &= J_n P^T A^T P J_n \\ &= P^T J_n A^T J_n P && \text{(move } J_n \text{ past } P \text{ and } P^T\text{)} \\ &= P^T AP && \text{(use the persymmetry of } A\text{)} \\ &= B. \end{aligned}$$

This is exactly the defining equation for B to be persymmetric.

Necessity: a single entry forces the rule

Write E_{uv} for the matrix whose only nonzero entry is a 1 at (u, v). For each i, the matrix $E_{i,r(i)}$ is persymmetric because its nonzero entry is on the anti-diagonal.

Let $q = p^{-1}$. Reordering by $P^T(\cdot)P$ moves this entry to $(q(i), q(r(i)))$. A matrix with just one nonzero entry can be persymmetric only if that entry is on the anti-diagonal. Otherwise its reflection is a different, zero entry. The universal hypothesis therefore requires

$$q(r(i)) = r(q(i)) \qquad \text{for every } i.$$

Substitute $i = p(k)$ and apply p to both sides. This gives $r(p(k)) = p(r(k))$, which is (2).

Finally, PJ and JP send e_i to $e_{p(r(i))}$ and $e_{r(p(i))}$, respectively, so the index rule is equivalent to $PJ = JP$. Multiplying $PJ = JP$ on the left by P^T gives $P^T JP = J$; multiplying the latter on the left by P gives the converse. This proves all four equivalences. □

5. How many permutations are there?

Let $m = \lfloor n/2 \rfloor$. There are m mirror pairs. Choose an ordering of those pairs in $m!$ ways, then choose the orientation of each pair in 2^m ways. When n is odd, the middle index has no choice: it stays fixed. Consequently,

$$\boxed{\#\{P : \ P^T AP \text{ preserves persymmetry for every } A\} = 2^{\lfloor n/2 \rfloor} \lfloor n/2 \rfloor !.}$$

Dimension n	All permutations $n!$	Permitted permutations
1	1	1
2	2	2
3	6	2
4	24	8
5	120	8
6	720	48
7	5040	48

In dimension 2, both permutations work. In dimension 3, the middle index must stay fixed, and the only possibilities are I_3 and J_3.

All eight orderings in dimension 4

Under our convention, the permitted lists are

$$\begin{array}{llll} (1,2,3,4), & (1,3,2,4), & (2,1,4,3), & (2,4,1,3), \\ (3,1,4,2), & (3,4,1,2), & (4,2,3,1), & (4,3,2,1). \end{array}$$

Notice that not every valid permutation is its own inverse. For $p = (2,4,1,3)$,

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

which commutes with J_4 but satisfies $P^T \neq P$. Thus replacing P^T by P in a calculation is not justified in general.

6. Why this matters

Eigenvalues and nilpotency are preserved

Since $P^{-1} = P^T$, the matrices A and $B = P^T AP$ are similar. For example,

$$\begin{aligned} \det(xI - B) &= \det\big(P^T(xI - A)P\big) \\ &= \det(P^T)\det(xI - A)\det(P) \\ &= \det(xI - A). \end{aligned}$$

They therefore have the same characteristic polynomial and eigenvalues, including multiplicities. Also,

$$B^2 = P^T AP\, P^T AP = P^T A^2 P, \qquad B^k = P^T A^k P.$$

Thus $A^k = 0$ if and only if $B^k = 0$. For instance,

$$U = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad J_3^T U J_3 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Both matrices are persymmetric, have characteristic polynomial x^3, and have cube zero but square nonzero. The upper-shift and lower-shift patterns represent the same case under an admissible permutation.